

Requirements Table — Digital Campus Library Reservation Gateway

Lab 1: Requirements Engineering & UML Use-Case Modelling | Problem Statement #08 — Campus & Academic Operations

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow members to search the ISBN repository and place hold requests on currently loaned books to enter a FIFO waitlist.	High	Pass: User is assigned correct position in queue. Fail: A user already holding the maximum allowed books is able to place an additional reservation.	Core reservation functionality; guarantees fair, first-come-first-served access to high-demand titles.
FR-002	Functional	The system shall notify a student member that their reserved book is available for pickup within 5 minutes of the book being checked in and marked available.	High	Pass: The member receives a notification within 5 minutes of the book's check-in timestamp for 95% of test cases. Fail: Notification is delayed beyond 5 minutes or not delivered.	Prevents queue stagnation and wasted holding-shelf space by ensuring the member acts on the hold promptly. (Delivery channel and mechanism are left to the design phase.)
FR-003	Functional	The system shall automatically calculate overdue fines at a rate of ₹5 per day per book, capped at ₹200 per book, starting the day after the due date.	Medium	Pass: Calculated fine matches the formula exactly for test cases at 1, 10, 40, and 60+ days overdue (cap applied). Fail: Any test case returns an incorrect fine amount.	Ensures consistent, transparent, and auditable fine tracking without manual librarian calculation.
FR-004	Functional	The system shall allow a student member to cancel their own pending hold request at any time before the reserved copy is checked out to them.	Medium	Pass: Cancelled hold is removed from the queue immediately and the next member in line advances one position. Fail: Cancelled hold remains visible in the active queue.	Keeps the FIFO queue accurate and frees reserved copies for other waiting members as early as possible.
FR-005	Functional	The system shall allow the Head Librarian to manually reorder or remove entries in a book's holding queue for exceptional cases (e.g., a lost or damaged copy).	Low	Pass: The queue reflects the librarian's manual change and logs the librarian ID, timestamp, and reason. Fail: The change is applied without being logged, or is not applied at all.	Provides administrative flexibility to handle real-world exceptions that a purely automated FIFO queue cannot resolve.
NFR-001	Nonfunctional (Performance & Security)	The catalog search query shall return paginated results for 50,000 titles in less than 150 ms.	High	Pass: Benchmarking tests confirm target latency and security standards are met under simulated peak load. Fail: p95 latency exceeds 150 ms or security checks fail under load.	Search is the entry point to every workflow; slow or insecure search degrades the experience for the entire student body.
NFR-002	Nonfunctional (Reliability & Availability)	The system shall maintain at least 99.5% uptime during official library operating hours (8:00 AM–10:00 PM), excluding pre-announced maintenance windows.	High	Pass: Monitoring logs show uptime $\geq 99.5\%$ over a rolling 30-day period. Fail: Uptime falls below 99.5% during operating hours in any 30-day window.	Students depend on the gateway during open hours for time-sensitive holds and fine payments; downtime directly blocks core library operations.